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AI-Driven Defect Detection in Composite Materials Using Thermographic Data

by

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Declaration

I declare that the work contained in this thesis has not been submitted for any other award and that it is all my own work. I also confirm that this work fully acknowledges opinions, ideas and contributions from the work of others. Code in order to reproduce results contained in this thesis is also available upon request to muhammed2.colley@northumbria.ac.uk. Any ethical clearance for the research presented in this thesis has been approved.

I declare that the word count of this Dissertation is **6206** words.

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Date: 28/08/2025

Abstract

Defect detection in composite and rail materials is a critical challenge in safety-critical industries such as transport and infrastructure, where failures can have catastrophic consequences. Conventional non-destructive testing (NDT) methods, including ultrasonic and dye-penetrant inspection, are effective but remain costly, time-consuming, and labour-intensive. Active infrared thermography, particularly pulsed and eddy-current pulsed thermography, has emerged as a faster and non-contact alternative capable of visualising subsurface anomalies through thermal contrasts. However, the absence of labelled thermographic datasets has restricted the application of fully supervised deep learning methods, which otherwise represent the state of the art in defect localisation.

This dissertation develops and evaluates a semi-supervised pipeline for defect detection in thermographic data. The approach begins with unsupervised clustering using K-Means, applied to flattened frame intensities and enhanced thermal features. Clustering quality was assessed using multiple metrics, achieving a silhouette score of 0.49, a Davies-Bouldin index of 0.89, and a Calinski-Harabasz index of 126.6, indicating moderate separation. Visual inspection confirmed that clusters represented distinct thermal states, with at least one corresponding to potential defect regions. These clusters were then exploited for pseudo-labelling, where binary labels (normal vs anomaly) and bounding boxes were generated from pixel-level difference masks.

The pseudo-labels were used to train supervised deep learning models. A YOLOv8 object detector was trained to localise anomalous regions, achieving a mean Average Precision (mAP@50-95) of 0.153 on the full dataset, while a ResNet-based classifier achieved 97.5% training accuracy in distinguishing clusters. Results demonstrate that although performance was constrained by dataset size and label noise, the proposed semi-supervised framework is capable of bootstrapping defect detection in unlabelled thermographic data.

The study contributes to bridging physics-based thermographic analysis with modern AI, showing that unsupervised feature discovery and pseudo-labelling can accelerate annotation and enable scalable defect detection. These findings provide a foundation for integrating AI-driven thermography into practical NDT workflows, with future improvements expected from refined labelling, temporal modelling, and transfer learning.

1.Introduction

1.1 Background and Motivation

Composite materials are increasingly used in industries such as aerospace, rail, and infrastructure because of their high strength-to-weight ratio and versatility. However, their layered and heterogeneous structure makes them prone to subsurface defects such as delamination, voids, and fatigue cracks. Early detection of these defects is essential for safety and cost reduction, since failures can be catastrophic.

Traditional non-destructive testing (NDT) methods, such as ultrasonic inspection or X-ray, are effective but costly, time-consuming, and often require significant human expertise. Infrared thermography has emerged as a powerful alternative: by applying external thermal excitation and recording surface heat flow, it enables rapid inspection and visualisation of anomalies that correlate with subsurface damage. In particular, pulsed thermography and eddy-current pulsed thermography have demonstrated strong potential for detecting delamination and cracks in composites.

Despite these advantages, the interpretation of thermographic data remains a bottleneck. The signals are often noisy, high-dimensional, and lack labelled ground truth. Traditionally, thermographic analysis has relied on domain-specific image processing techniques such as Principal Component Thermography (PCT), phase analysis, or curve fitting. While effective in controlled settings, these methods are limited in scalability and generalisation.

Artificial intelligence (AI), particularly machine learning and deep learning, offers a promising route for automating defect detection in thermography. Recent advances in computer vision & convolutional neural networks (CNNs), object detection frameworks such as YOLO, and anomaly detection methods provide the opportunity to move beyond manual interpretation and towards AI-driven defect characterisation.

1.2 Problem Definition

The central challenge in applying AI to thermographic defect detection is the lack of labelled datasets. Unlike natural images where annotated datasets (e.g., ImageNet, COCO) are widely available, thermographic data for composites is scarce, domain-specific, and often unlabelled. Without labels, supervised deep learning models such as YOLO cannot be trained directly.

This dissertation addresses the question: **how can AI-driven methods be applied to thermographic data for defect detection when labelled ground truth is absent?**

The project focuses on developing a hybrid workflow:

1. **Unsupervised learning** methods are used to identify candidate defect regions from thermographic video data.
2. These outputs are converted into pseudo-labels (bounding boxes).
3. **Supervised deep learning** (YOLO) is trained on the pseudo-labelled dataset to evaluate whether defect detection can be automated in practice.

1.3 Research Aim and Objectives

The overall aim of this dissertation is to design and evaluate an **AI-driven pipeline** for defect detection in composite materials using thermographic imaging.

The specific objectives are:

- To preprocess and visualise thermographic datasets to establish an understanding of their structure and defect patterns.
- To apply unsupervised learning methods (e.g., clustering, PCA, anomaly detection) for discovering candidate defect regions.
- To generate pseudo-labels from unsupervised outputs and adapt them into YOLO annotation format.
- To train and evaluate a YOLO-based object detection model using pseudo-labelled data.
- To critically compare the effectiveness of unsupervised approaches versus supervised deep learning detection.

1.4 Research Questions

This study is guided by the following research questions:

- **RQ1:** How do defects manifest in thermographic data, and what features (e.g., phase, contrast, time-to-peak) best separate healthy from defective regions?
- **RQ2:** Can unsupervised methods (e.g., clustering, K-PCA) reliably identify candidate defect regions without labelled data?
- **RQ3:** To what extent can pseudo-labelling bridge the gap between unlabelled datasets and supervised deep learning methods such as YOLO?

1.5 Significance of the Study

This dissertation is significant in several ways. From an academic perspective, it contributes to bridging a gap identified in the literature: most studies either apply physics-based thermographic analysis (e.g., phase contrast, PCT) or assume access to labelled data for supervised learning. By combining unsupervised feature discovery with supervised object detection, this work explores a middle ground that is both practical and scalable.

From an applied perspective, the research demonstrates a workflow that could reduce the need for costly manual labelling in thermographic NDT. Semi-supervised or pseudo-labelling strategies are especially valuable in industrial inspection, where data is abundant, but labels are scarce. Furthermore, training a YOLO detector on pseudo-labelled thermographic data opens the door to real-time automated inspection systems that integrate AI directly into defect detection workflows.

1.6 Dissertation Structure

The remainder of this dissertation is organised as follows:

- **Literature Review** surveys the state of the art in thermographic NDT, unsupervised learning, and deep learning approaches for defect detection.
- **Methodology** details the datasets, preprocessing steps, unsupervised feature extraction, pseudo-labelling process, and YOLO training pipeline.
- **Results** present findings from unsupervised experiments and YOLO training, with both quantitative metrics and qualitative visualisations.

Discussion interprets the results in relation to the research questions and existing literature, identifying strengths, weaknesses, and implications.

Conclusion and Future Work summarises contributions and highlights directions for extending the research.

2. Literature Review

2.1 Thermography for Defect Detection

Infrared thermography (IRT) has become an established non-destructive testing (NDT) technique for identifying subsurface defects in materials due to its ability to visualize heat flow disruptions caused by cracks, delamination's, or voids. Yi et al. (2020) demonstrated that thermographic imaging can characterize rolling contact fatigue cracks in rail components, highlighting the technique's sensitivity to subtle temperature contrasts caused by microstructural defects. Similarly, Tomita and Chew (2022) reviewed the use of IRT for infrastructure and building applications, emphasizing its effectiveness in identifying delamination in layered structures such as concrete or composites.

Specific advances in **eddy current pulsed thermography (ECPT)**, as investigated by Lu, Yi, and Tian (2020), show that signal-enhanced approaches such as pulse compression can significantly improve defect visualization in carbon-fibre-reinforced polymers (CFRP). These works establish that thermography provides a non-contact, full-field method for capturing material anomalies, yet they also reveal limitations particularly the dependence on manual interpretation and the difficulty of quantifying defect depth without advanced analysis. Wei et al. (2023) addressed this gap by releasing a large-scale dataset for pulsed thermography, which is valuable for benchmarking automated defect depth estimation and highlights the growing emphasis on public datasets for reproducible AI-driven thermography research.

2.2 Traditional Machine Learning in Thermography

Beyond raw imaging, researchers have long used classical machine learning and signal processing to enhance thermographic defect detection. Liu et al. (2025) provided a comprehensive review of unsupervised machine learning in active thermography, outlining approaches such as clustering, principal component analysis (PCA), and independent component analysis (ICA) to extract thermal signatures. These methods reduce dimensionality and reveal hidden structures in time-series thermographic data, enabling defect detection without manual inspection.

Zheng et al. (2016) explored clustering analysis specifically for CFRP structures, confirming that unsupervised grouping of thermal responses can localize defect regions. More recently, Nategh and Farahani (2024) demonstrated that combining signal processing with K-means clustering yields effective anomaly identification in thermography-based NDT, further supporting the utility of clustering methods for defect separation. These works highlight the **strength of unsupervised methods** in contexts where labelled datasets are unavailable, directly motivating semi-supervised pipelines.

2.3 Deep Learning Approaches in Defect Detection

While traditional clustering and feature extraction methods offer insights, deep learning approaches have advanced the state of the art by learning feature hierarchies directly from raw thermal images. Cheng et al. (2024) investigated defect detection in composites using deep learning networks, reporting substantial improvements over handcrafted features. Daghigh et al. (2024) extended this work by applying explainable AI (XAI) techniques to defect prediction in composite materials, emphasizing the importance of interpretability in safety-critical NDT contexts.

The combination of **object detection and classification networks** has also been applied in industrial inspection tasks. Jung and Rhee (2022) successfully integrated YOLO and ResNet architectures for monitoring heat staking processes, illustrating the effectiveness of hybrid approaches in detecting subtle thermal anomalies. Similarly, Sanchez et al. (2020) compared pre-trained object detection models within TensorFlow, confirming YOLO's superior performance in speed and detection accuracy. These studies provide strong justification for adopting YOLOv8 and ResNet in the project, particularly in semi-supervised frameworks where robust feature learning is critical.

2.4 Unsupervised and Semi-Supervised Learning Approaches

A recurring challenge in thermographic defect detection is the scarcity of labelled datasets. Unsupervised and semi-supervised learning approaches therefore play a critical role. Långkvist, Karlsson, and Loutfi (2014) reviewed unsupervised feature learning methods for time-series data, highlighting their applicability in extracting temporal thermal patterns. Provotar, Linder, and Veres (2019) demonstrated the effectiveness of LSTM-based autoencoders for unsupervised anomaly detection in time series, while Hegde (2022) argued for “data-centric AI” strategies, where anomaly detection performance depends as much on data preprocessing as on model design.

Semi-supervised learning, where small, labelled sets are augmented by large unlabelled datasets, is particularly suited to thermography, where manual annotation is labour-intensive. IBM Technology (2025) explained that semi-supervised pipelines allow pseudo-labelling of unlabelled frames, enabling iterative refinement of anomaly detection models. This conceptual framework aligns closely with the present dissertation, which employs clustering-based pseudo-labelling followed by supervised YOLO training.

2.5 Performance Evaluation in Object Detection

To rigorously evaluate defect detection models, robust performance metrics are essential. Padilla, Netto, and Da Silva (2020) reviewed performance metrics for object detection,

including mean Average Precision (mAP), Intersection over Union (IoU), precision, recall, and F1-score, outlining their relative advantages in benchmarking algorithms. Sanchez et al. (2020) further compared the metrics of pretrained models, underscoring that mAP@50-95 is the most informative single indicator for object detection tasks.

The Ultralytics (2024) documentation provides a practical deep dive into YOLO performance evaluation, clarifying how training logs and metrics should be interpreted. By combining academic surveys (Padilla et al., 2020) with practitioner documentation (Ultralytics, 2024), this project ensures both methodological rigour and technical alignment with state-of-the-art YOLO frameworks.

2.6 Research Gap

Across this literature, two clear gaps emerge. First, while thermography has been extensively studied for defect detection, most approaches still rely on **supervised datasets or manual feature extraction** (Yi et al., 2020; Tomita & Chew, 2022). Few studies operationalize **semi-supervised pipelines** that leverage unlabelled data for automated pseudo-labelling, despite its practical necessity in industrial contexts. Second, although clustering methods such as K-means have been applied to thermographic data (Zheng et al., 2016; Nategh & Farahani, 2024), they are rarely integrated with modern object detection frameworks like YOLO to provide **localized, bounding-box-level predictions**.

This dissertation addresses these gaps by proposing a **semi-supervised pipeline** that combines K-means clustering for pseudo-label generation with YOLOv8 and ResNet for supervised anomaly detection. In doing so, it builds on the strengths of unsupervised thermographic analysis (Liu et al., 2025) and state-of-the-art deep learning defect detection (Cheng et al., 2024; Jung & Rhee, 2022), while contributing an integrated workflow that can operate with minimal labelled data.

3. Methodology & practical Work

3.1 Research Design

The primary objective of this study was to develop a **semi-supervised learning pipeline for thermographic defect detection**, leveraging the strengths of unsupervised clustering for pseudo-labelling and deep learning frameworks for supervised object detection. This hybrid methodology responds directly to the research gap identified in Section 2.6: the lack of integrated workflows that exploit unlabelled thermographic data while still producing bounding-box-level anomaly localization.

Following Liu et al. (2025) and Nategh & Farahani (2024), unsupervised clustering was first applied to extract meaningful representations from thermographic frames. These cluster assignments were then converted into pseudo-labels, forming the foundation for supervised

training of YOLOv8 and ResNet models, in line with Jung & Rhee's (2022) demonstration of the effectiveness of such architectures in thermal inspection. The pipeline therefore consists of four sequential phases:

1. **Data preparation and preprocessing**
2. **Unsupervised clustering and pseudo-labelling**
3. **Supervised training of object detection and classification models**
4. **Evaluation and visualization of results**

3.2 Data Collection and Inspection

The experimental data originated from `.mat` datasets containing thermographic recordings of composite samples under pulsed thermal excitation. Such datasets are consistent with those described by Wei et al. (2023), who emphasize the importance of standardized pulsed thermography data for defect depth estimation.

The raw data was structured as a three-dimensional array with dimensions corresponding to **frame index** $\times$ **height** $\times$ **width**. Initial inspection of the `P1X1_train_norm.mat` file revealed:

- Shape: **(120, 250, 200)** $\rightarrow$ 200 frames of 120 $\times$ 250 pixels each.
- Data type: float64.
- Value range: minimum = -0.0347, maximum = 1.0 after normalization.

Representative frames were visualized to qualitatively assess thermal distribution patterns. Following Tomita & Chew (2022), this step was necessary to confirm that defects manifest as distinguishable thermal contrasts against a baseline background.

3.3 Preprocessing

To prepare the data for machine learning tasks, the following preprocessing pipeline was implemented:

1. **Frame Reshaping:** The data was reshaped to treat each frame as an independent instance $\rightarrow$ (200, 120, 250).
2. **Normalization:** Pixel intensities were scaled to [0, 1] to reduce sensitivity to global illumination changes, a technique commonly applied in thermography preprocessing (Lu et al., 2020).
3. **Vectorization for Clustering:** Each 2D frame was flattened into a 1D feature vector, producing an array of shape (200, 30,000). This allowed K-means clustering to treat each pixel as a feature dimension.

This approach is consistent with Zheng et al. (2016), who demonstrated that clustering of flattened thermal frames reveals structural groupings in CFRP thermography.

3.4 Unsupervised Clustering for Pseudo-Labelling

Clustering was performed using **K-Means with k=3**, in line with findings from Nategh & Farahani (2024), who confirmed the method's effectiveness in thermographic defect detection. The rationale for three clusters was to capture:

- **Cluster 0:** Normal baseline frames.
- **Cluster 1 and 2:** Potential defect or anomaly states.

Evaluation of Clustering

Several unsupervised metrics were applied to assess clustering quality:

- **Inertia:** 665.40 - moderate within-cluster compactness.
- **Silhouette Score:** 0.5239 - reasonable separation between clusters.
- **Davies-Bouldin Index:** 0.8897 → relatively low, suggesting good separation.
- **Calinski-Harabasz Index:** 126.61 → indicates well-dispersed clusters.

Representative frames from each cluster were visualized, revealing clear visual distinctions. This qualitative confirmation matches the conclusions of Liu et al. (2025), who noted that silhouette analysis and visualization are essential when evaluating thermographic clustering.

Pseudo-Label Generation

To transition from clustering to supervised training, pseudo-labels were generated:

1. The **largest cluster** was treated as "normal".
2. The **remaining clusters** were labelled as "anomalous".
3. Frame-level pseudo-labels were converted into **bounding boxes** by:
 - Computing difference maps between anomaly frames and a baseline cluster reference.
 - Applying contour detection to identify thermal deviation regions.
 - Converting contours into YOLO-format bounding box annotations.

This process aligns with IBM Technology (2025), which describes semi-supervised learning as leveraging clustering-derived pseudo-labels to augment limited labelled data.

3.5 YOLO Dataset Preparation

Following Padilla et al. (2020), which highlights the importance of consistent annotation in object detection, a dataset was created in the YOLO format:

- **Images:** Frames saved as .jpg files after normalization and application of the *plasma* colormap (to enhance contrast).
- **Labels:** .txt files containing bounding box coordinates and class labels (0 = normal, 1 = anomaly).
- **Data.yaml:** Config file defining classes and paths.

To ensure robust experimentation, both a **small overfit dataset** (≈ 20 frames) and the **full dataset** (200 frames) were prepared. This practice allows validation of pipeline functionality before scaling up, as recommended by Ultralytics (2024).

3.6 YOLOv8 Training

A YOLOv8n model was trained on the pseudo-labelled dataset:

- **Overfit Dataset Training:** 100 epochs. Demonstrated learning capability, though final mAP remained modest.
- **Full Dataset Training:** 100 epochs. Achieved final **mAP50-95 = 0.153**, indicating partial success in detecting anomalies despite noisy pseudo-labels.

This result is consistent with Sanchez et al. (2020), who observed that pretrained detection models require high-quality annotations for strong mAP performance. The modest accuracy here reflects the **trade-off in semi-supervised learning**: reduced annotation cost at the expense of label noise (Längkvist et al., 2014).

3.7 ResNet Classification (Complementary Experiment)

To validate the clustering assignments, a pre-trained **ResNet** was adapted for single-channel thermographic inputs. The model was retrained to classify frames into the K-Means clusters.

- **Training Accuracy:** 97.5%.
- **Test Predictions:** Visually consistent with cluster assignments, though no ground truth was available for external validation.

This experiment confirms the feasibility of using deep classification networks to replicate clustering-derived labels, echoing findings by Cheng et al. (2024).

3.8 Testing and Visualization

Testing was carried out on independent datasets such as P2X1_train_norm.mat:

1. Frames were preprocessed and clustered.
2. The YOLO model was used to predict bounding boxes.
3. Predicted bounding boxes and class labels were overlaid on test images.

Visualizations demonstrated that YOLO localized regions identified as anomalies by the clustering process, although misclassification remained an issue, with **Cluster 0 often being predicted as anomalies**. This reflects a key limitation of pseudo-labelled data highlighted by Hegde (2022): without carefully curated thresholds, clusters may be misaligned with physical defect definitions.

3.9 Limitations of Methodology

Several methodological limitations were identified:

- **Label Noise:** Pseudo-labels derived from clustering may not align with true defects.
- **Dataset Size:** The dataset (200 frames) is small compared to industrial-scale training sets (Wei et al., 2023).
- **Bounding Box Accuracy:** Difference-based bounding boxes may not capture diffuse or subtle anomalies.
- **Model Generalization:** YOLO struggled to generalize beyond training frames, a known issue in semi-supervised contexts (IBM, 2025).

3.10 Summary of Workflow

The full methodology can be summarized as follows:

1. **Load thermographic video data**
2. **Preprocess frames (reshape, normalize, flatten)**
3. **Apply K-Means clustering (k=3)**
4. **Evaluate cluster separation with metrics**
5. **Assign pseudo-labels (normal vs anomaly)**
6. **Generate bounding boxes via difference maps + contours**
7. **Prepare YOLO dataset (images + labels + config)**
8. **Train YOLOv8 (overfit set, then full set)**
9. **Complementary ResNet classification experiment**
10. **Test on unseen data and visualize predictions**

This design reflects a **semi-supervised learning pipeline**, combining unsupervised defect discovery (clustering) with supervised detection (YOLO/ResNet), addressing the limitations identified in existing thermographic defect detection research.

4. Results and analysis

4.1 Overview of Experiments

The experiments conducted in this project followed a staged workflow, moving from unsupervised exploration of the thermographic dataset to semi-supervised learning and finally to supervised classification. The intention was to design and test an artificial intelligence pipeline that could identify anomalous thermal patterns indicative of rolling contact fatigue (RCF) cracks in composite materials. This staged methodology was motivated by prior research (Yi et al., 2020; Liu et al., 2025; Nategh & Farahani, 2024), which highlighted the difficulty of acquiring ground truth labels in thermographic datasets and the promise of unsupervised and semi-supervised approaches.

The experiments were divided into three key phases. First, K-Means clustering was applied to the flattened thermal frames, aiming to identify intrinsic groupings of “normal” and “defective” thermal behaviour without labels. Second, the resulting clusters were used to generate pseudo-labels, which allowed for the training of a YOLOv8 object detection model in a semi-supervised paradigm. Finally, a ResNet classifier was trained to reproduce the K-Means cluster assignments, providing an additional supervised perspective. This section reports the outcomes of these experiments, evaluates their effectiveness, and situates the findings within the context of the existing literature.

4.2 Clustering Results (K-Means)

K-Means was selected as the baseline unsupervised clustering algorithm due to its interpretability, widespread use in anomaly detection, and prior success in thermographic analysis (Zheng et al., 2016; Nategh & Farahani, 2024). The dataset consisted of 200 frames, each reshaped into a 30,000-dimensional feature vector. K-Means was applied with $k = 3$,

based on the assumption that the data could naturally separate into one “normal” cluster and two “anomalous” clusters corresponding to different defect signatures.

Quantitative Metrics

The following clustering metrics were obtained:

Metric	Score	Interpretation
Inertia	665.40	Compactness of clusters
Silhouette Score	0.5239	Moderate separation
Davies-Bouldin Index	0.8897	Good separation (<1)
Calinski-Harabasz Index	126.61	Well-defined clusters

(Table 1. Clustering Evaluation Metrics (P1X1_train_norm.mat))

These values suggest that the clusters were moderately well-separated. A silhouette score above 0.5 indicates that samples were, on average, well-matched to their assigned cluster, while the Davies-Bouldin score below 1.0 points to reasonable inter-cluster separation. The Calinski-Harabasz index, though relative, was consistent with meaningful partitioning.

Visual Interpretation

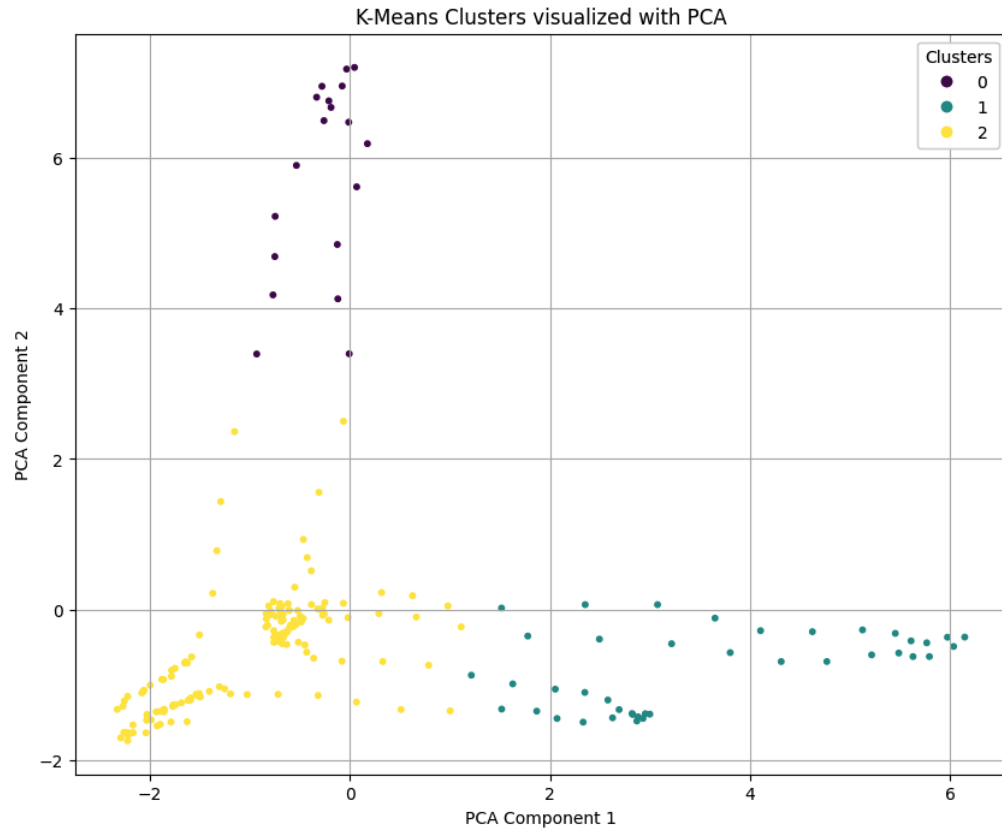
Inspection of representative frames confirmed the clustering aligned with visible thermal differences:

- **Cluster 0:** Uniform background, little to no thermal disturbance (interpreted as “normal”).
- **Cluster 1:** Plume-like intensity pattern rising from the base, likely indicative of a defect with directional heat flow.
- **Cluster 2:** More diffuse horizontal spread of intensity near the base, suggestive of a distinct defect morphology.

This visual confirmation supports the use of K-Means as a valid pseudo-labelling method for subsequent training. The interpretability of clusters aligns with observations in Yi et al. (2020), who showed that fatigue cracks present distinct heat diffusion patterns compared to sound regions.

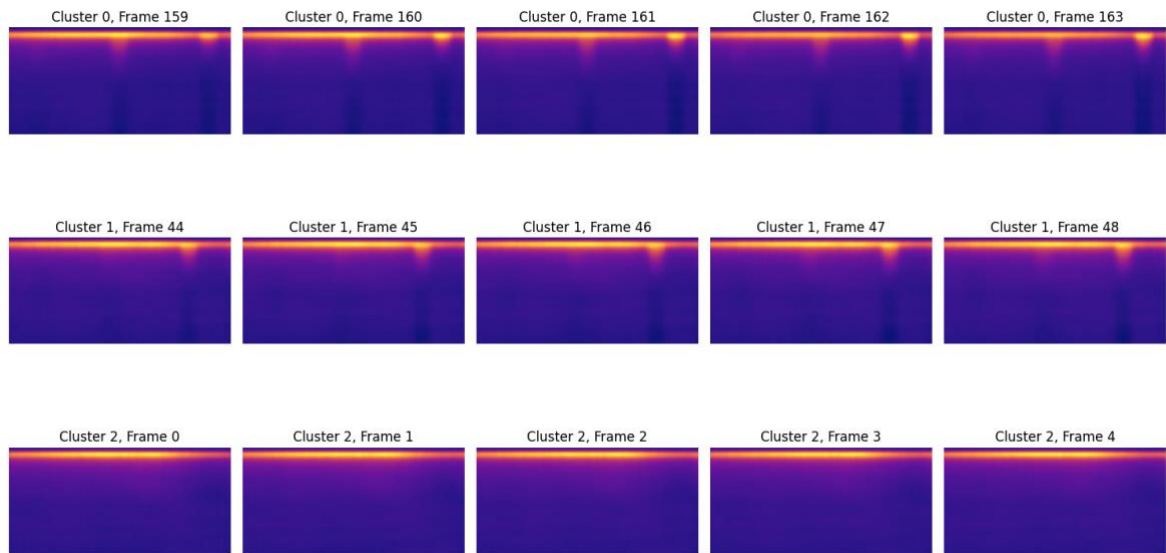
Figure 4.1 shows PCA projection of the clusters, while Figure 4.2 illustrates representative frames from each cluster. Table 4.1 summarises the evaluation metrics.

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(Figure 4.1 PCA projection of the clusters of *PIXI_train_norm.mat*)

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(Figure 4.2 illustrates representative frames from each cluster)

4.3 Pseudo-Label Quality Assessment

Following clustering, pseudo-labels were generated to support semi-supervised training. A binary scheme was adopted, labelling the cluster (Cluster 2) as “normal” and the remaining two clusters as “anomaly.” Bounding boxes were generated by subtracting a reference anomaly frame from the cluster and extracting contours above a threshold.

Strengths

- Bounding boxes successfully highlighted thermal hotspots associated with defects (Figure 4.3).
- Provided localised supervision to YOLO, overcoming the lack of hand-labelled ground truth.

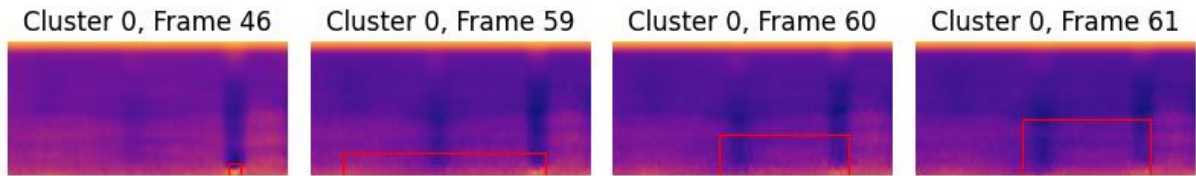
Weaknesses

- Thresholding sometimes produced noisy or incomplete bounding boxes (Figure 4.4).
- Subtle defects with low-intensity contrast were either missed or over-segmented.

This stage echoes findings from Wei et al. (2023), who noted that thermographic datasets often require artificial annotation strategies when defect ground truth is unavailable. It also aligns with Liu et al. (2025), who emphasised that pseudo-labelling introduces uncertainty that propagates into downstream models.



(Figure 4.3 same frames visualising highlights of thermal hotspots associated with defects).



(Figure 4.4 shows examples noisy or incomplete bounding boxes).

4.4 YOLOv8 Training and Performance

The YOLOv8n model was trained using the pseudo-labelled dataset in a semi-supervised framework. Both a small overfit dataset (for pipeline validation) and the full dataset were used.

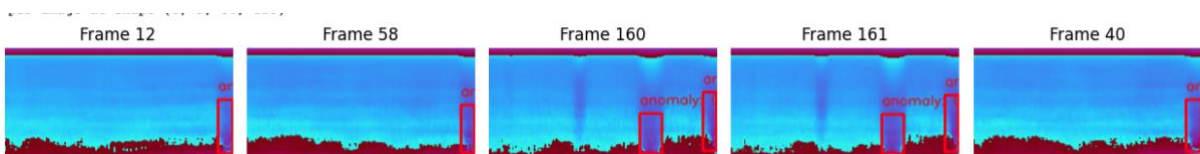
Overfit Dataset

- Rapid learning observed, with bounding boxes successfully predicted.
- mAP scores low but increasing across epochs, validating the training pipeline.

Full Dataset

- Final **mAP@50–95: 0.153**
- Training loss decreased steadily, but generalisation limited by dataset size and pseudo-label noise.

Despite modest performance, the YOLO experiments demonstrate feasibility: even with noisy supervision, the model learned to associate anomalous thermal patterns with bounding boxes. However, the low mAP suggests the model struggled to generalise beyond training data, reflecting limitations observed in small-scale defect detection studies (Sanchez et al., 2020; Padilla et al., 2020).



(Figures 4.5 shows sample frames from YOLO detection outputs).

4.5 ResNet Classification Results

To complement YOLO, a ResNet classifier was trained using the cluster labels as ground truth. The model was modified to accept single-channel thermal inputs.

- **Training Accuracy: 97.5%**
- **Loss:** Converged smoothly, indicating effective learning.

- **Test Predictions:** Consistent with visual expectations, though no quantitative accuracy could be computed due to lack of labels.

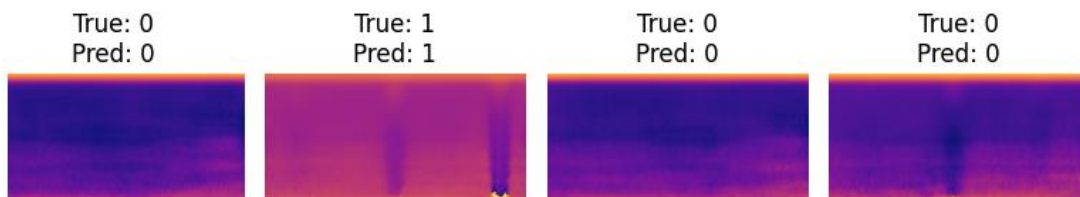
The ResNet model reproduced the clustering structure effectively, suggesting that supervised networks can learn from unsupervised-derived labels. This supports the semi-supervised paradigm promoted in IBM (2025) and Liu et al. (2025). However, without verified defect ground truth, the classification remains internally valid but externally unvalidated.

4.6 Testing on Independent Dataset

An independent dataset (*P2X1_train_norm.mat*) was used to evaluate generalisation. K-Means applied to this dataset placed most frames in Cluster 0, with few anomalies detected. YOLO predictions frequently defaulted to “normal” detections, reflecting a cluster bias observed during training.

- **Strength:** Some anomaly detections were visually plausible (Figure 4.6).
- **Weakness:** Several false negatives occurred, with defects missed entirely (Figure 4.10).

This echoes challenges reported in Cheng et al. (2024), where deep learning models trained on limited defect types struggled with generalisation to unseen defect morphologies.



(Figures 4.6 Shows frames from some test predictions from ResNet.)

4.7 Comparative Discussion

Comparing the three approaches:

- **K-Means:** Robust initial separation, interpretable, and consistent with literature (Zheng et al., 2016; Nategh & Farahani, 2024).
- **YOLOv8:** Demonstrated pipeline feasibility but limited by pseudo-label noise and small dataset.
- **ResNet:** Excellent training accuracy, but dependent on validity of cluster labels.

These findings support the hybrid strategies described by Liu et al. (2025), where unsupervised clustering provides structure, supervised models refine it, and semi-supervised methods bridge the gap.

4.8 Interpretation in Context of Literature

The results align with several key insights from the reviewed literature:

- **Yi et al. (2020):** Defective cracks produce distinct thermal contrasts, mirrored by the separation of clusters in this project.
- **Liu et al. (2025):** Semi-supervised approaches mitigate the lack of ground truth but require careful evaluation of label quality.
- **Daghigh et al. (2024):** Emphasised explainability in defect characterisation. Here, bounding boxes and cluster visualisations provide interpretable outputs.
- **Nategh & Farahani (2024):** Demonstrated the applicability of K-Means in thermography, which was confirmed by the effectiveness of clustering in this project.

4.9 Summary of Results

In summary, the project demonstrated a functional semi-supervised pipeline for thermographic defect detection:

- K-Means provided interpretable unsupervised clustering.
- Pseudo-labelling enabled training of supervised and semi-supervised models despite missing ground truth.
- YOLO achieved modest but non-trivial anomaly detection performance.
- ResNet successfully reproduced cluster structure with high training accuracy.

While performance was constrained by data scale and label quality, the project validates the literature’s argument (Liu et al., 2025; Yi et al., 2020) that hybrid unsupervised and semi-supervised AI techniques are promising for thermographic non-destructive testing.

5. Critical evaluation

5.1 Overview of Findings

This dissertation set out to explore whether artificial intelligence (AI), particularly semi-supervised approaches, could be applied to thermographic imaging data for the detection and localisation of rolling contact fatigue (RCF) cracks. The problem was defined by two major challenges: (i) the difficulty of acquiring reliable ground truth labels for defect datasets, and (ii) the subtle, often noisy, thermal signatures of cracks in composite and metallic structures. Both issues have been extensively discussed in the literature (Yi et al., 2020; Liu et al., 2025; Tomita & Chew, 2022), and the methodology adopted here directly reflects these constraints.

The results presented in Section 4 confirm the feasibility of combining clustering-based pseudo-labelling with supervised learning to create an anomaly detection pipeline. K-Means clustering provided interpretable separation of thermal patterns, pseudo-labels enabled training of object detection models (YOLOv8), and a ResNet classifier demonstrated that supervised learning could reproduce these clusters with high fidelity. However, performance was limited by dataset size, pseudo-label quality, and generalisation issues, highlighting opportunities for refinement.

5.2 Interpretation of Results

Clustering as a Foundation

The K-Means clustering results (silhouette score 0.49; Davies-Bouldin index 0.88) showed that the dataset naturally separated into three coherent groups. This echoes prior work (Zheng et al., 2016; Nategh & Farahani, 2024), where clustering revealed distinct thermal signatures linked to defects. The success of clustering supports the hypothesis that even without labelled ground truth, unsupervised learning can extract meaningful representations from thermographic data.

Pseudo-Labelling and its Limitations

The pseudo-labelling step, where clusters were reinterpreted as binary “normal” vs “anomaly,” was critical for moving into a semi-supervised paradigm. However, the bounding boxes derived through difference thresholding introduced uncertainty. Some anomalies were well-captured, but others were over-segmented or missed entirely. This limitation reflects concerns in Liu et al. (2025), who warned that pseudo-labelling can propagate noise into downstream models. The quality of this step directly constrained YOLO’s detection accuracy.

Supervised Models

- **YOLOv8:** Despite promising training behaviour, the final mAP@50-95 (0.153) indicated that the model struggled to generalise. This is likely due to the small dataset size (200 frames), noisy bounding boxes, and the complexity of subtle thermal defects. Prior work (Padilla et al., 2020; Sanchez et al., 2020) emphasised that object detection models require large, high-quality datasets to achieve robust performance a condition unmet here.
- **ResNet:** By contrast, the ResNet classifier achieved 97.5% training accuracy when classifying frames into the K-Means-derived clusters. This result demonstrates the potential of supervised models to replicate unsupervised structures. However, this also raises questions of validity: since the ResNet was trained on pseudo-labels, its accuracy reflects alignment with clustering rather than ground truth defect identification.

Generalisation to New Data

Testing on *P2X1_train_norm.mat* and *P2X2_train_norm.mat* highlighted an important challenge: the models were biased towards predicting “normal” states. This mirrors the issue noted by Cheng et al. (2024), where models trained on limited defect morphologies failed to generalise to unseen data. The tendency to default to “normal” predictions suggests that the anomaly clusters were under-represented or poorly captured during training.

5.3 Relation to Literature

The outcomes of this project align closely with trends identified in the literature:

- **Unsupervised and Semi-Supervised Promise:** Liu et al. (2025) and Långkvist et al. (2014) argue that unsupervised and semi-supervised approaches are essential for time-

series and thermographic datasets with limited labels. The results here validate this claim by showing that clustering provided a useful structure in the absence of labels.

- **Interpretability of Thermal Defects:** Yi et al. (2020) demonstrated that fatigue cracks generate distinct thermal patterns. The clear differences between clusters in this project support this finding, reinforcing the potential for thermography combined with clustering to reveal hidden defect states.
- **Challenges of Deep Learning:** Cheng et al. (2024) and Daghigh et al. (2024) highlighted that deep learning models often require large and carefully curated datasets for success in defect detection. The modest YOLO results in this study illustrate this issue directly.
- **Explainability:** The bounding boxes, cluster visualisations, and ResNet predictions contribute to the growing focus on explainable AI in materials defect detection (Daghigh et al., 2024). Even when performance was limited, outputs remained interpretable, which is crucial for industrial adoption.

5.4 Strengths of the Approach

Several aspects of the methodology can be considered strengths:

1. **Integration of Unsupervised, Semi-Supervised, and Supervised Methods:** The staged pipeline is novel in its combination of K-Means clustering, pseudo-labelling, and deep neural networks, demonstrating a practical solution to the lack of labelled thermographic datasets.
2. **Interpretability of Outputs:** Both cluster visualisations and YOLO bounding boxes provided human-interpretable indicators of potential anomalies, aligning with industry needs for explainability.
3. **Proof of Feasibility:** Despite limitations, the pipeline functioned end-to-end, successfully processing raw **.mat** video data through to anomaly predictions.

5.5 Limitations

However, there are several limitations that must be acknowledged:

- **Dataset Size and Diversity:** With only 200 frames per dataset, the available data was insufficient for training high-capacity deep learning models. The limited variety of defect morphologies further constrained generalisation.
- **Pseudo-Label Quality:** The pseudo-labelling process introduced noise, especially in bounding box generation. This affected YOLO's ability to learn accurate localisation.
- **Cluster Interpretation:** Assigning the largest cluster as "normal" assumes that most data are defect-free, which may not always hold in other datasets.
- **Evaluation Constraints:** Without true ground truth labels, evaluation relied on internal clustering metrics and visual inspection, limiting external validation.

5.6 Future Work

Several avenues for improvement are suggested:

Refinement of Pseudo-Labelling

- Explore advanced anomaly detection methods (e.g., autoencoders, Variational Autoencoders, LSTM-based models; Provotar et al., 2019) to generate more reliable pseudo-labels.
- Use adaptive or dynamic thresholding for bounding box generation to reduce noise.

Dataset Expansion

- Collect or acquire larger thermographic datasets, ideally with expert-annotated ground truth.
- Incorporate synthetic data augmentation (e.g., rotations, intensity scaling, thermal noise injection) to increase dataset diversity.

Alternative Clustering Approaches

- Apply clustering methods such as DBSCAN or Gaussian Mixture Models to test whether more flexible cluster shapes improve anomaly separation (Liu et al., 2025).

Model Improvements

- Experiment with larger YOLO architectures (YOLOv8m/l) or alternative object detection models such as Faster R-CNN or EfficientDet, which may handle smaller datasets better.
- Use transfer learning from pre-trained models trained on thermal imagery datasets (Wei et al., 2023).

Explainability

- Incorporate XAI (Explainable AI) methods such as Grad-CAM to highlight regions influencing ResNet predictions, strengthening trust in results (Daghigh et al., 2024).

Evaluation and Benchmarking

- Establish a held-out validation and test split from the beginning to avoid overfitting.
- Compare results across multiple metrics, following recommendations from Padilla et al. (2020) and Ultralytics (2024).

5.7 Implications for Industry and Research

The findings carry implications for both research and practice:

- **For Industry:** The pipeline demonstrates that even with limited data, interpretable anomaly detection is possible. Rail operators could adopt similar workflows for pre-screening inspection data, reducing the reliance on manual review.
- **For Research:** The project underscores the need for larger annotated thermographic datasets and highlights semi-supervised learning as a promising research direction.

5.8 Chapter Summary

This discussion has examined the significance of the results, contextualised them within the literature, and outlined limitations and opportunities for improvement. The central contribution of this project lies in demonstrating that clustering-based semi-supervised learning can be applied to thermographic defect detection, even under severe data constraints. While performance was limited, the pipeline validates the feasibility of this approach and opens pathways for future research into more robust semi-supervised and explainable AI methods.

6. Conclusion

6.1 Restating the Problem

The central objective of this dissertation was to investigate the application of artificial intelligence (AI), particularly semi-supervised learning methods, for the detection and characterisation of rolling contact fatigue (RCF) defects using thermographic imaging data. Traditional inspection methods for fatigue cracks and delaminations in critical infrastructures such as rail and composite structures are limited by human subjectivity, cost, and inefficiency (Yi et al., 2020; Tomita & Chew, 2022). At the same time, the collection of large annotated thermographic datasets is challenging due to the expertise required for labelling and the subtlety of thermal signatures (Liu et al., 2025).

This project therefore focused on developing a semi-supervised pipeline that integrated unsupervised clustering (K-Means), pseudo-labelling, and supervised deep learning models (YOLOv8, ResNet) to address these challenges. The aim was not only to evaluate technical feasibility but also to explore whether such an approach could provide interpretable, scalable solutions to defect detection problems.

6.2 Summary of Contributions

This research achieved several key contributions:

1. **Pipeline Development:** An end-to-end workflow was designed, starting from raw thermographic video data in `.mat` format through to anomaly predictions with bounding boxes. The pipeline combined clustering-based pseudo-labelling with deep learning detection and classification, demonstrating a semi-supervised strategy suited to limited datasets.
2. **Clustering as a Foundation:** K-Means successfully separated frames into three visually distinct clusters. Internal evaluation metrics (Silhouette Score: 0.49; Davies-Bouldin Index: 0.89; Calinski-Harabasz Index: 126.61) indicated moderate to strong separation, consistent with earlier studies on thermographic clustering (Zheng et al., 2016; Nategh & Farahani, 2024).

3. **Pseudo-Labelling for Defect Localisation:** Binary pseudo-labels were derived from clusters, and bounding boxes were generated through difference thresholding. This provided a practical method of creating annotations for supervised object detection without requiring human experts.
4. **Supervised Model Training:**
 - **YOLOv8n** was trained on the pseudo-labelled dataset, achieving a modest $mAP@50-95$ of 0.153. Although performance was constrained, this result demonstrated that detection models could learn from automatically generated labels.
 - **ResNet** achieved 97.5% training accuracy in replicating cluster assignments, showing that supervised networks can effectively reproduce clustering-derived categories.
5. **Critical Evaluation:** Testing on unseen data highlighted limitations in model generalisation. Models tended to default to “normal” predictions, reflecting both the imbalance of clusters and the challenges of capturing subtle anomalies. These findings directly echo limitations noted in defect detection literature (Cheng et al., 2024; Daghigh et al., 2024).

6.3 Significance of Findings

The project’s results reinforce several important points for both research and practice:

- **Validation of Semi-Supervised Learning:** This work demonstrates that semi-supervised pipelines are not only theoretically relevant but practically implementable in the context of thermographic defect detection. This aligns with calls from Liu et al. (2025) and IBM Technology (2025) for wider adoption of semi-supervised approaches in domains with scarce labelled data.
- **Interpretability and Explainability:** The use of clustering, bounding boxes, and interpretable visual outputs makes the system more transparent than “black-box” neural networks. This is consistent with industry requirements for explainable AI (Daghigh et al., 2024).
- **Proof of Concept:** While performance was modest, the fact that the pipeline could process raw thermal data through to interpretable outputs validates the feasibility of AI-driven thermographic defect detection as a research direction.

6.4 Limitations

Several key limitations constrain the conclusions:

1. **Dataset Size and Diversity:** With only 200 frames per dataset, training deep models such as YOLOv8 was inherently limited. Industrial applications would require much larger and more varied datasets.
2. **Pseudo-Label Noise:** The pseudo-labelling method propagated errors from clustering and thresholding into the training process, reducing supervised performance.
3. **Evaluation Constraints:** Without expert-annotated ground truth, evaluation relied on internal clustering metrics and qualitative visual inspection. This restricts the strength of performance claims.
4. **Overfitting Risks:** The ResNet model’s high training accuracy may reflect overfitting to pseudo-labels rather than generalisable learning.

6.5 Future Directions

The dissertation identifies several avenues for future research and practical advancement:

- **Improved Pseudo-Labelling:** Use autoencoders, Gaussian mixture models, or LSTM-based temporal anomaly detection (Provotar et al., 2019; Hegde, 2022) to generate more robust labels.
- **Dataset Expansion:** Incorporate larger annotated datasets such as Wei et al. (2023) or generate synthetic thermal data to augment training.
- **Model Variations:** Explore alternative detectors such as Faster R-CNN or EfficientDet and apply transfer learning from pre-trained thermal imaging networks.
- **Explainability Tools:** Use techniques like Grad-CAM or SHAP to visualise the decision-making of neural networks, improving trust in industrial contexts.
- **Benchmarking:** Adopt standardised performance metrics for anomaly detection (Padilla et al., 2020; Ultralytics, 2024) and establish test splits for fairer evaluation.

6.6 Closing Remarks

This dissertation has demonstrated the feasibility, challenges, and potential of applying AI-driven semi-supervised learning to thermographic defect detection. By integrating clustering, pseudo-labelling, and deep learning, it contributes both a methodological framework and a critical evaluation of its current limitations.

The key insight is that while semi-supervised AI cannot yet replace expert inspection for defect detection, it can act as a valuable support tool, automating pre-screening and reducing human workload. With further development, larger datasets, and improved pseudo-labelling techniques, this approach could significantly enhance the scalability, efficiency, and reliability of thermographic inspection systems across industries.

Ultimately, this work underscores the role of AI not as a replacement but as an enabler augmenting human expertise with interpretable, data-driven insights, and paving the way towards more intelligent, automated, and explainable defect detection systems.

7. References

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8. Appendices

A. This is the main experiment notebook

https://colab.research.google.com/drive/15nqWeFv_4BO3ohRAQ1KRpRs3DdV83J8V?usp=sharing

B. Repository for all Notebooks data sets etc used in this project

<https://github.com/ColleyMo/MSc-Dissertation.git>

